

Practical - 8

Practical Title: MySQL Schema Design, ER Model, PDO Connectivity, and Prepared Statements

Problem Definition: Design the database for StudentHub with students, events and registrations. Create PHP database connection using MYSQL statements.

A. Theory

1. Is the schema normalized properly?

Answer: Yes, the schema is normalized properly with no redundancy.

2. Are primary keys and foreign keys correctly defined?

Answer: Yes, all primary and foreign key constraints are correctly linked.

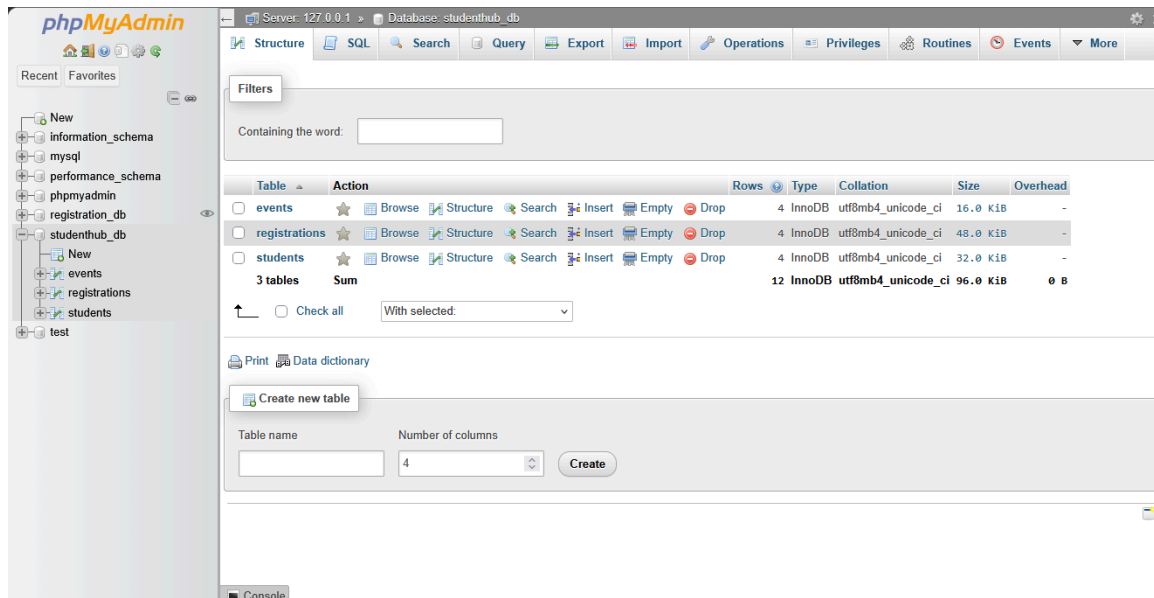
3. Is the database connection handled securely?

Answer: Yes, the database connection is handled securely using PDO exceptions and prepared statements.

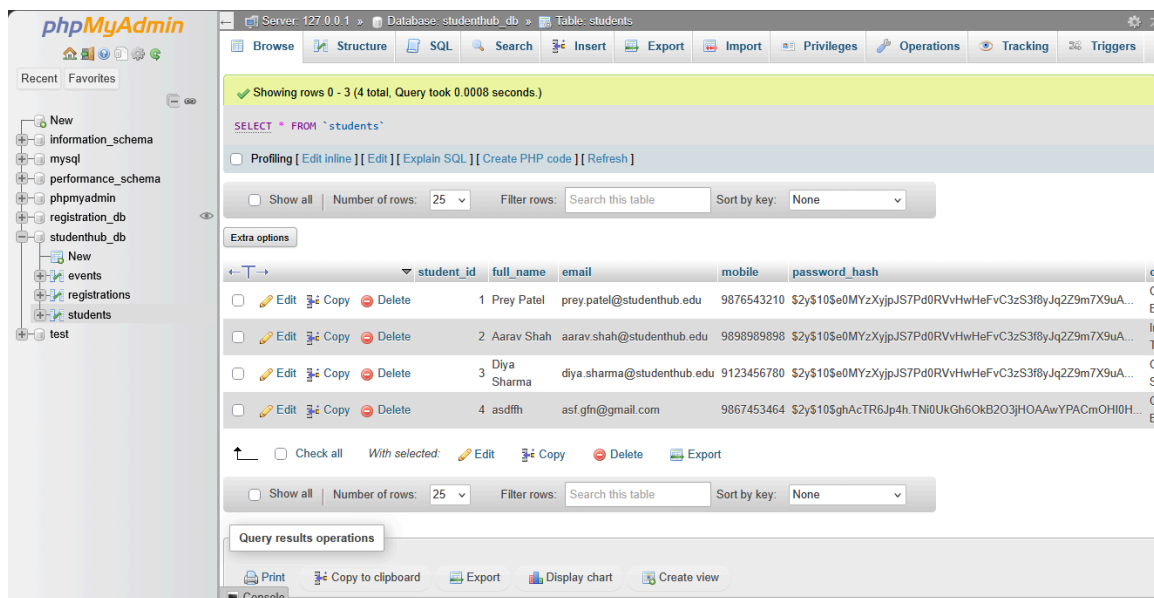
B. Implementation/ Output/Result

Purpose	What it does
Database Design	Created 3 tables: students, events and registrations with foreign keys.
Connection	Connects PHP to MySQL using PDO.
Queries	Inserts and reads data using prepared statements to prevent SQL injection.

C. Evidence of Work (Screenshots)



Description: Figure 8.1 database and its tables



Description: Figure 8.2 Students table

Server: 127.0.0.1 > Database: studenthub_db > Table: registrations

Showing rows 0 - 3 (4 total, Query took 0.0006 seconds.)

```
SELECT * FROM `registrations`
```

Number of rows: 25 | Filter rows: Search this table | Sort by key: None

	registration_id	student_id	event_id	status	registered_at
☐ Edit ☐ Copy ☐ Delete	1	1	1	Confirmed	2026-10-03 22:27:27
☐ Edit ☐ Copy ☐ Delete	2	2	1	Confirmed	2026-10-03 22:27:27
☐ Edit ☐ Copy ☐ Delete	3	2	1	Confirmed	2026-10-03 22:27:27
☐ Edit ☐ Copy ☐ Delete	4	3	3	Confirmed	2026-10-03 22:27:27

Check all | With selected: Edit Copy Delete Export

Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations: Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Description: Figure 8.3 Registrations table

Server: 127.0.0.1 > Database: studenthub_db > Table: events

Showing rows 0 - 3 (4 total, Query took 0.0004 seconds.)

```
SELECT * FROM `events`
```

Number of rows: 25 | Filter rows: Search this table | Sort by key: None

	event_id	title	category	event_date	venue	max_seats	created_at
☐ Edit ☐ Copy ☐ Delete	1	Web Development Hackathon 2026	Technical	2026-10-15	Main Computer Lab	60	2026-10-03 22:27:26
☐ Edit ☐ Copy ☐ Delete	2	Cloud & DevOps Bootcamp	Workshop	2026-10-22	Seminar Hall B	45	2026-10-03 22:27:26
☐ Edit ☐ Copy ☐ Delete	3	Tech Career Fair & Mock Interviews	Career	2026-11-05	Auditorium 1	100	2026-10-03 22:27:26
☐ Edit ☐ Copy ☐ Delete	4	Annual Cultural Fest - Resonance	Cultural	2026-11-20	College Open Ground	250	2026-10-03 22:27:26

Check all | With selected: Edit Copy Delete Export

Number of rows: 25 | Filter rows: Search this table | Sort by key: None

Query results operations: Print Copy to clipboard Export Display chart Create view

Bookmark this SQL query

Console

Description: Figure 8.4 Events table

D. GitHub Repository Link:Link: <https://github.com/prep04-yl/StudentHub-Portal>**E. Challenges Faced:**

Understanding foreign key constraints and debugging PDO connection errors.

F. Learning Outcome:

I learned how to design and connect a normalized database securely using PHP.

Faculty Remarks: (Keep this blank for faculty)